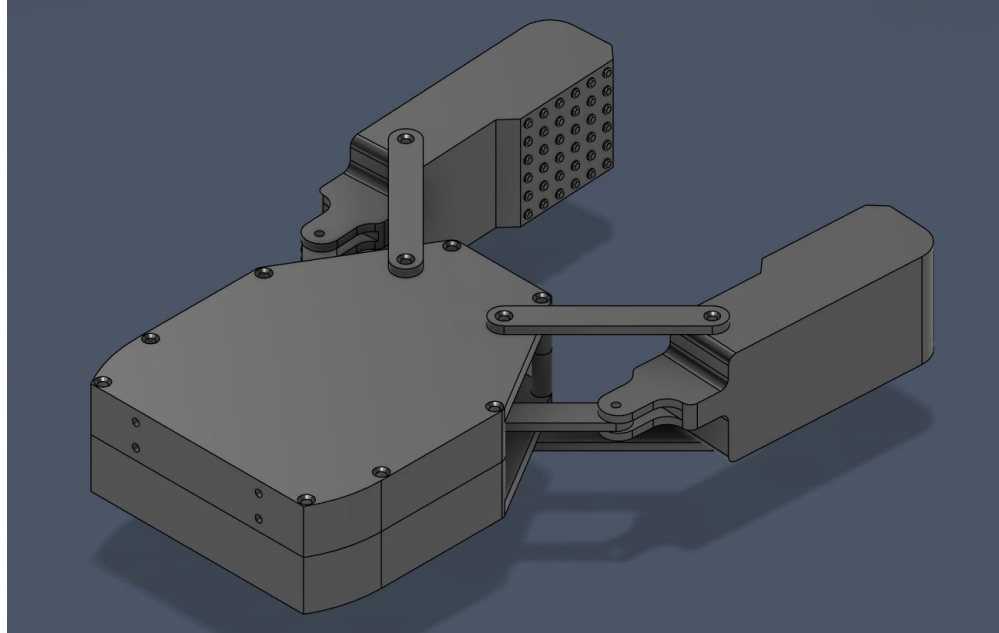


Design and Development of a Parallel 4-Bar Linkage Gripper



Project Objectives

The Objective:

To design and prototype a gripper capable of carrying a payload of 500g with a torque safety margin

Design Requirements and Constraints:

Payload Target	500g vertical lift and horizontal movement
Kinematic Constraint	Parallel finger motion (4-bar linkage)
Torque Requirement	Achieve a 4x SF
Actuation	Integration with a standard high-torque servo and microcontroller

Kinematic Design

Mechanism: Parallel 4-Bar Linkage

Why: Ensures that the gripping force vector remains perpendicular to the object surface throughout the entire stroke.

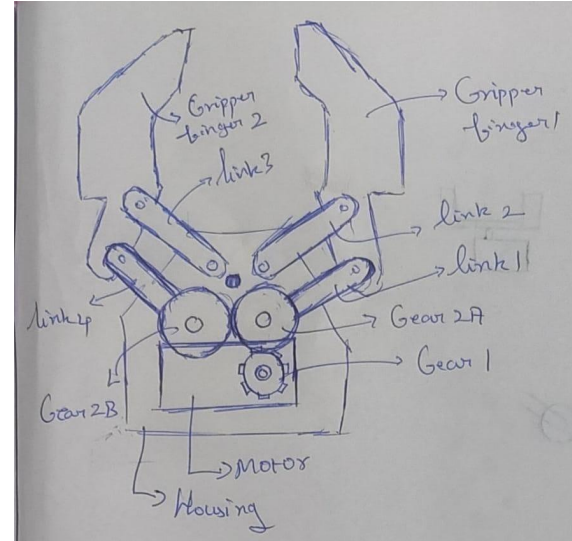
DOF: 1

Linkage Type: Parallelogram Linkage

Stroke Parameters:

Max Opening – 100 mm

Min Opening – 35mm



Gear Train

Gear Specification:

Module (m): 0.5

Driven Gears (x2): $N = 72$, $D = 36\text{mm}$

Driver Gears (x1): $N = 24$, $D = 12\text{mm}$

Centre-to-Centre Distance (C):

V1: Driver & Driven - Theoretical = 24mm

- Actual = 24.25mm

Both Driven - Theoretical = 36mm

- Actual = 36mm

V2: Need to add few mm to actual for driven for DFM purposes

Reduction Ratio 3:1

Load Analysis & Torque Budgeting

Theoretical Output:

Servo Stall Torque: 20 kgcm

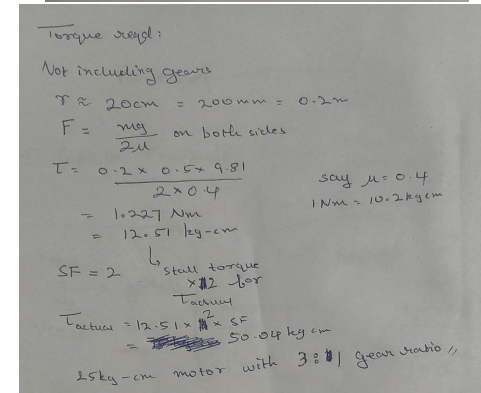
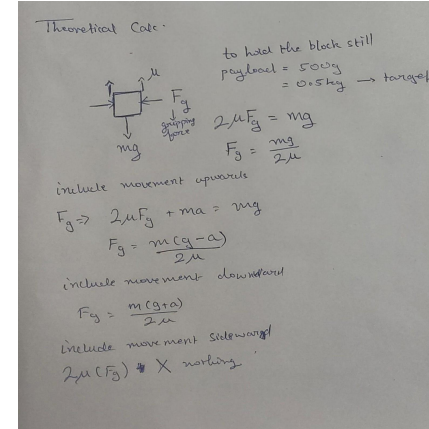
Transmission Output: 60 kgcm

Required Torque: 12.51 kgcm

Safety Factor: 4

Actual Torque Required: 50.04 kgcm

By designing for 60 kgcm, the system effectively treats the 500 g payload as a 2 kg equivalent load, ensuring the motor never reaches its mechanical stall limit during operation.



BOM & Hardware

1x MG958 Motor: Selected for its 20 kgcm stall torque

2x MR105 Bearings: To support the 72T gears

6x M3 Nuts: Secure the attachment of 2 parts together by bolts

8x M3 Bolts (30mm length): For bolting the top and bottom chassis together

10x M3 Bolts (12-15mm length): For bolting the links to the chassis and fingers

4x M3 Bolts (5mm length): For bolting 2 of the links to the 72T gears

18x Washers: For tightening and friction reduction for revolution of links

1x Arduino Uno: Microcontroller/ Brain of the project

3x Gears: To ensure 60 kgcm torque is reached by using 3:1 gear ratio

Housing: To hold all parts within the gripper system and to give it a nice look

6x Links: Part of the 4-bar linkage system for parallel gripping

2x Gripper Fingers: Used as fingers to grip onto objects

Assembly Overview

Chassis: Both the chassis are bolted together by 8 M3 bolts (30 mm long)

Top and Bottom Links:

Chassis – bolt head -> washer -> link -> washer -> chassis -> washer -> nut

Finger – bolted with 12-15mm long M3 bolts

Middle Links:

Gears – bolted with 5mm-long M3 bolts

Finger – bolted with 12-15mm long M3 bolts

Gears:

72T – MR105 bearing snapfit to top chassis, gear snap fit to bearing

24T – Directly bolted to MG958 motor

Motor: Bolted to bottom chassis with M3 bolts

V1 vs. V2: Lessons in DFM

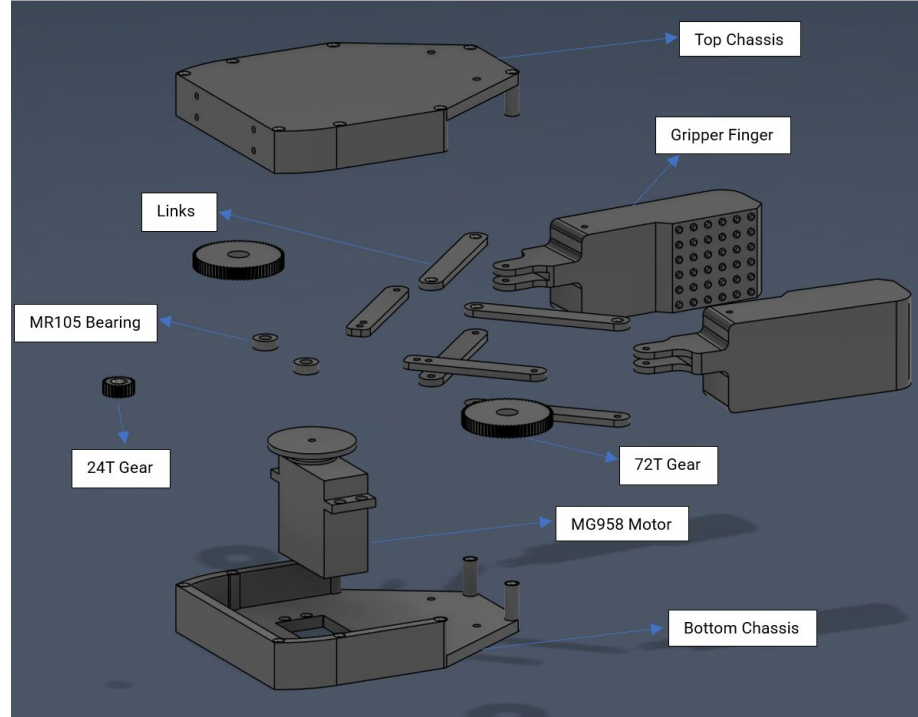
V1 Problems:

- In V1, M3 bolt holes were undersized due to 3D printing
- The bearing housing was incorrectly dimensioned, leading to a loose
- Flat edges made it difficult to align gears and bolts during manual assembly
- Both Driven Gears were exactly 36 mm apart, thus making it difficult for the gears to rotate

V2 Fix:

- Increased the hole size and added 1mm chamfers to all the holes in the design
- Increased the 36 mm distance to 36.25 mm distance
- Used Vernier Calliper to check exact dimensions and fixed bearing housing diameter

Exploded View



THANK YOU!